

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6564**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600017
		Course Section:	1		

L61_474_986_1_1505_48173



Name of the candidate. Dip Nilesh Jain

Candidate's UID number:

2	0	2	2	6	0	0	0	1	7
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Examination class room number:

2	0	2
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Invigilator's signature with date:

 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE

(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday, 8/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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1. a

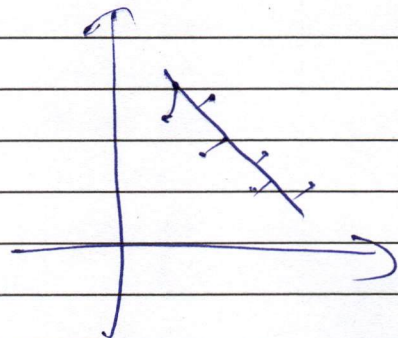
$$f(w) = \frac{1}{2} \|Xw - y\|_2^2$$

- gradient descent is very useful method used to reduce cost/loss function.
- It modifies the weight/parameter in opposite direction to their gradient.

$$\Delta \theta_n = \theta_n - \eta \Delta \theta$$

- it will try to minimize

$$\hat{y} - y = \hat{y} - (mx_i + c_i)$$



$$mse \Rightarrow \frac{1}{n} \sum (\hat{y} - (mx_i + c_i))^2$$

We need ^{value of} ~~minimize~~ m, c such that it gets minimized.

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2.9. - for 2D convolution ~~between~~ for

Input Image $\rightarrow H \times W \times c$

filter $\rightarrow f \times f$

k filters

~~Computational~~ complex

In convolution, filter is applied on image as sliding window, in each patch it makes element wise multiplication and add all to generate a new value

Convolution $\rightarrow (H \times W) * (f \times f) \rightarrow (H-f+1) \times (W-f+1)$

assuming 0 padding and 1 stride

and to each convolution it requires $(f \times f)$ multiplication

So computational complexity $\rightarrow (H-f+1) \times (W-f+1) \times (f \times f)$

but for depthwise separable convolution, we require

$\rightarrow$

$(H \times W \times c) * (f \times f \times k) \rightarrow (H-f+1) \times (W-f+1) \times (c-f+1)$

and for each convolution it requires $f \times f \times k$ computation

So computational

complexity $\rightarrow (H-f+1) \times (W-f+1) \times (c-f+1) \times (f \times f \times k)$

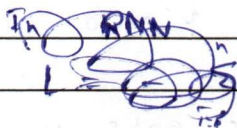


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	2.b	• Autoencoders are the types of neural network tend to learn the data in compressed form. • Autoencoders are consist of two parts encoder and decoder. • encoder converts the data into lower dimensional form keeping its important features intact • decoder converts lower dimensional data to its actual form • there are two types of autoencoders undercomplete autoencoder and overcomplete autoencoder. • Undercomplete autoencoder converts the original data into lower dimensional data. • So model forced to learn important features to keep it possible to get original data back • So a linear undercomplete autoencoder trained with MSE learns a subspace equivalent to PCA. • The overcomplete autoencoder tends to convert data to higher dimension. • It doesn't learn anything, just learn to copy-paste the data • To make it learn important features we use regularization • There are some regularization methods like denoising and sparse method which are used. • In denoising while the transforming image from original to high dimension, this method remove the gaussian noise so the image will have more of the important feature. • In sparse method we reduce weights of unimportant part of data, so it get less weightage and occupy less part of higher dimensional image.

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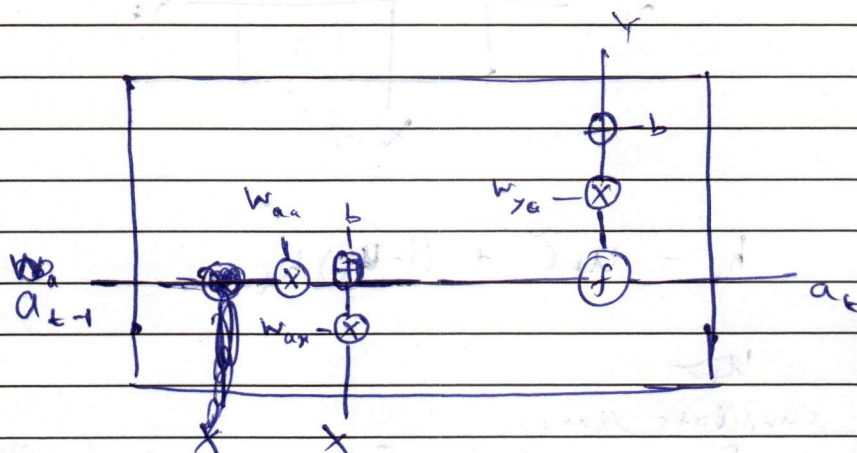
3.a



RNN →

$$a_t = \text{func}(W_{ax}x + W_{ah}a_{t-1} + b)$$

$$Y = \text{Softmax}(W_{ya}a_t + b_y)$$



$$\frac{\partial L}{\partial h_{t-k}} = \left(\frac{\partial L}{\partial a_t} \frac{\partial a_t}{\partial h} \frac{\partial h}{\partial h_{t-k}} \right)$$

As the t increases, the term gets shifted and tends to 0, so RNN loses memory. This problem is called vanishing gradient problem.

To solve this, LSTM has two states, memory state C_t keep important data from past timestamps, and remove/forget the unnecessary data using forget gate.

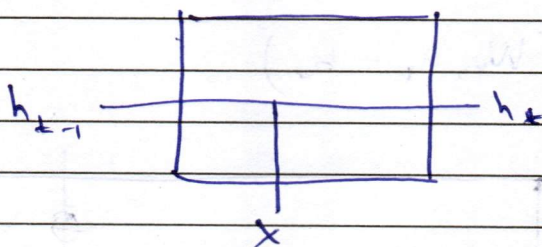
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3.6. GRU (gated recurrent unit) is a kind of lighter version of LSTM

- GRU have two gates, update gate and reset gate
- unlike LSTM it has only one hidden state, which is why it is less computation heavy



$$h_t = u_t \tilde{C}_t + (1 - u_t) h_{t-1}$$

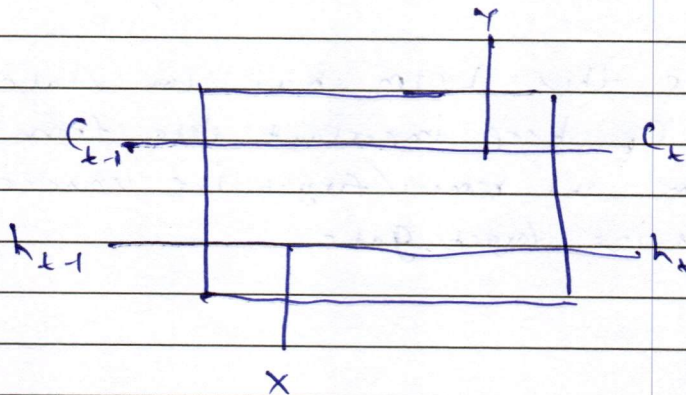
~~updates~~
candidate states

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x, v_t] + b_c)$$

$$r_t = \sigma(W_r[h_{t-1}, x] + b_r)$$

$$u_t = \sigma(W_u[h_{t-1}, x] + b_u)$$

• While in LSTM we have



$$h_t = i_t \tilde{C}_t + f_t h_{t-1}$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x] + b_c)$$

$$i_t = \sigma(W_i[h_{t-1}, x] + b_i)$$

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$$f_t = \sigma(w_f[h_{t-1}, x] + b_f)$$

$$o_t = \sigma(w_o[h_{t-1}, x] + b_o)$$

$$\tilde{r} = \tanh(C_t) \cdot o_t$$

- LSTM have 3 gates, input, output and forget gate.
- It maintains 2 states, C_t for long term memory and h_t for short term memory

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